AUTOMATING EMPLOYEE DATA

2.2 SOLUTION REQUIREMENTS

2.2.1 Functional Requirements

1. **Employee Registration**
   - The system should allow adding new employee details such as name, employee ID, department, designation, salary, and contact information.

2. **Employee Data Management**
   - The system should allow users to **view, update, and delete employee records** when necessary.

3. **Automated Data Update**
   - The system should automatically update employee information such as salary changes, department transfers, and promotions.

4. **Attendance Management**
   - The system should record and store employee attendance details.

5. **Payroll Processing**
   - The system should automatically calculate employee salaries based on attendance, allowances, and deductions.

6. **Report Generation**
    - The system should generate reports such as:
    - Employee list report
    - Salary report
    - Attendance report

7. **Search Functionality**
    - Users should be able to search employee details using employee ID, name, or department.

8. **User Authentication**
    - The system should allow only authorized users (admin/HR) to access and manage employee data.

9. **Data Validation**
    - The system should check and validate employee information to avoid incorrect or duplicate data.

10. **Data Storage**

- The system should store employee information securely in a database.

2.2.2 Non-Functional Requirements

Non-functional requirements describe **how the system performs its functions**.

1. **Performance**
   - The system should process employee data quickly and generate reports without delay.

2. **Security**
   - Only authorized users (Admin/HR) should be able to access or modify employee data.

3. **Reliability**
   - The system should store employee data safely and prevent data loss.

4. **Usability**
   - The system interface should be simple and easy to use for HR staff.

5. **Scalability**
   - The system should support a large number of employees as the organization grows.

6. **Data Backup**
   - The system should regularly back up employee records to avoid data loss.